



**Analyzing Single-Cell
Combinatorial Indexing RNA
Sequencing Data from Various Brain
Tumors to Identify Cell Types Present**

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PROJECT BACKGROUND

We used sci-RNAseq3 data from day 115 in fetal cerebrum development which was previously annotated. We used Seurat in R to compare this data to Medulloblastoma, Atypical Teratoid Rhabdoid Tumor, and Diffuse Intrinsic Pontine Glioma tumors data.

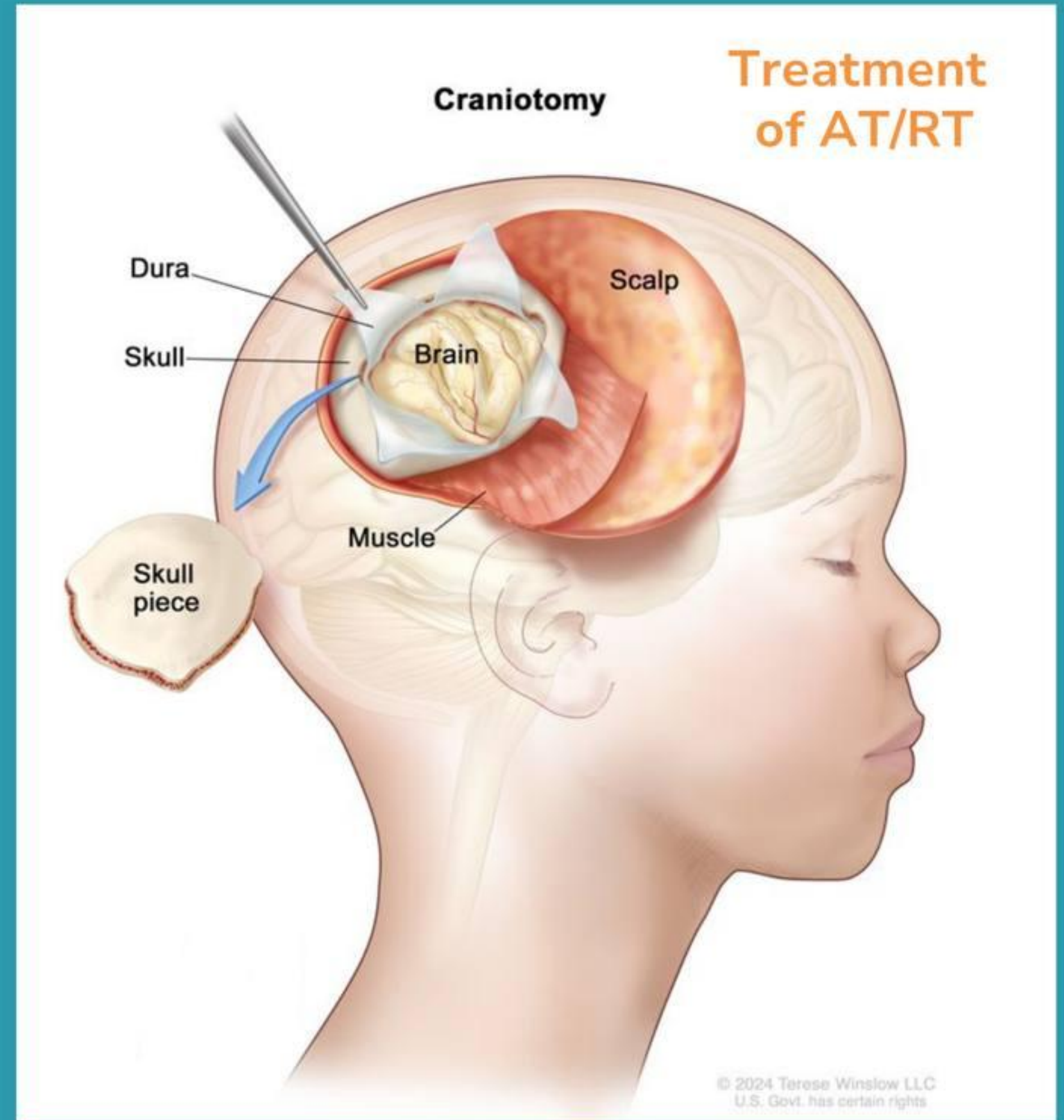
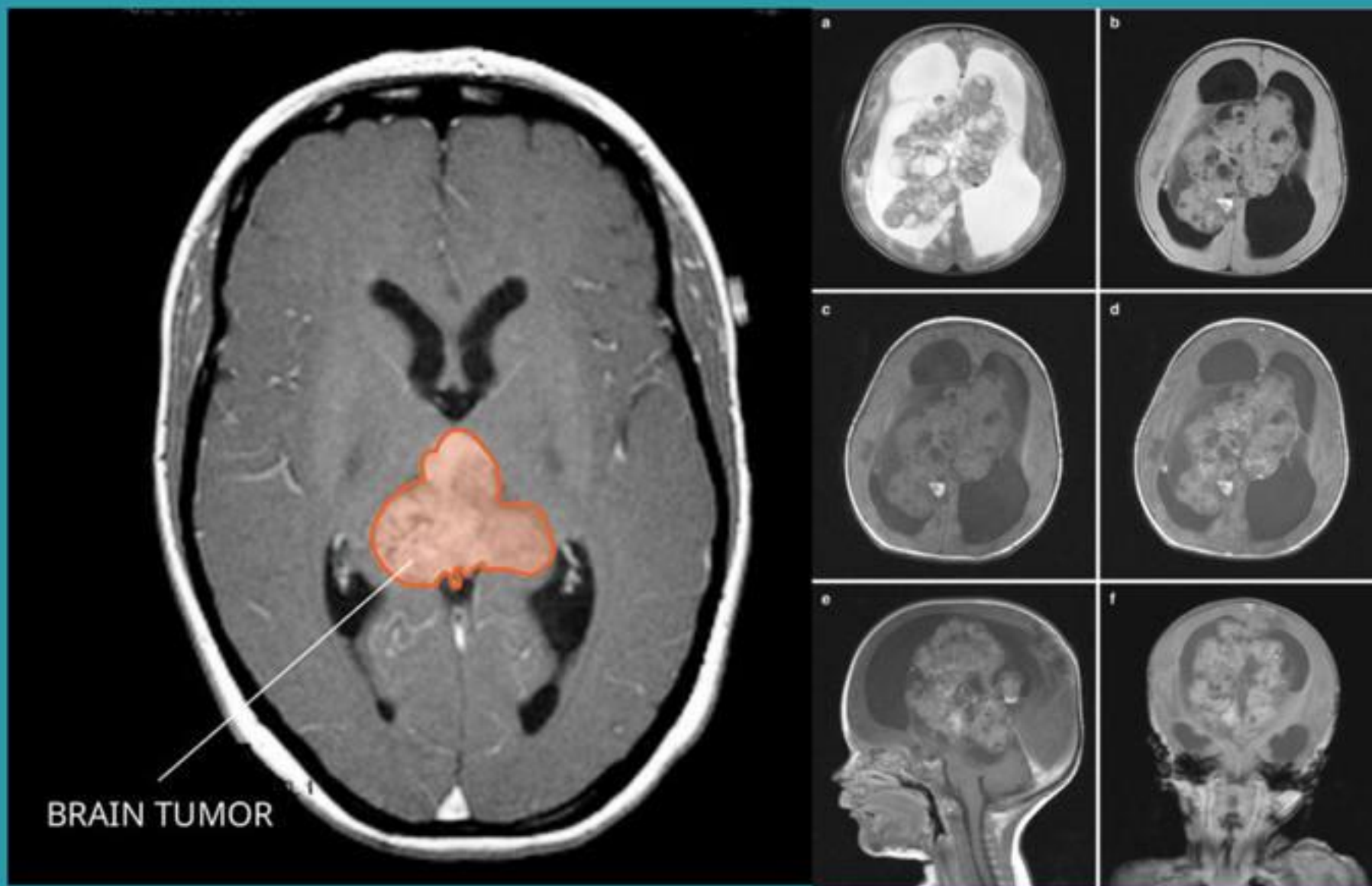
RESEARCH GOAL:

To have a better understanding of the three cancers and our current theories on why/how they develop

CANCER RESEARCH QUESTIONS

- Cell of origin
- Find the incidence and the survival rates
- What are the subgroups of this cancer?
- How were these subgroups identified?
- Is the cell of origin known/suspected for all groups of this cancer?
- Where (anatomical region) is it suspected to originate from?

AT/RT: Atypical Teratoid Rhabdoid Tumor



AT/RT: Atypical Teratoid Rhabdoid Tumor

- Cell of Origin: embryonal cells, including rhabdoid, neuroepithelial, epithelial, and mesenchymal cells
- Incidence:
 - Very rare, ~1-2% of all childhood brain tumors
 - Most commonly diagnosed in children under 3 y.o.
 - In the US, ~470 people live with AT/RT, only 50 adults
- Survival Rates:
 - 5-year relative survival rate low due to aggressiveness of tumor

AT/RT: Atypical Teratoid Rhabdoid Tumor

- Identified Subgroups:
 - AT/RT-TYR: More common in younger children.
 - AT/RT-SHH: Associated with the sonic hedgehog signaling pathway.
 - AT/RT-MYC: Most frequent in adults.
- Identification of Subgroups:
 - Included whole-genome sequencing, RNA sequencing, and comparative genomic hybridization

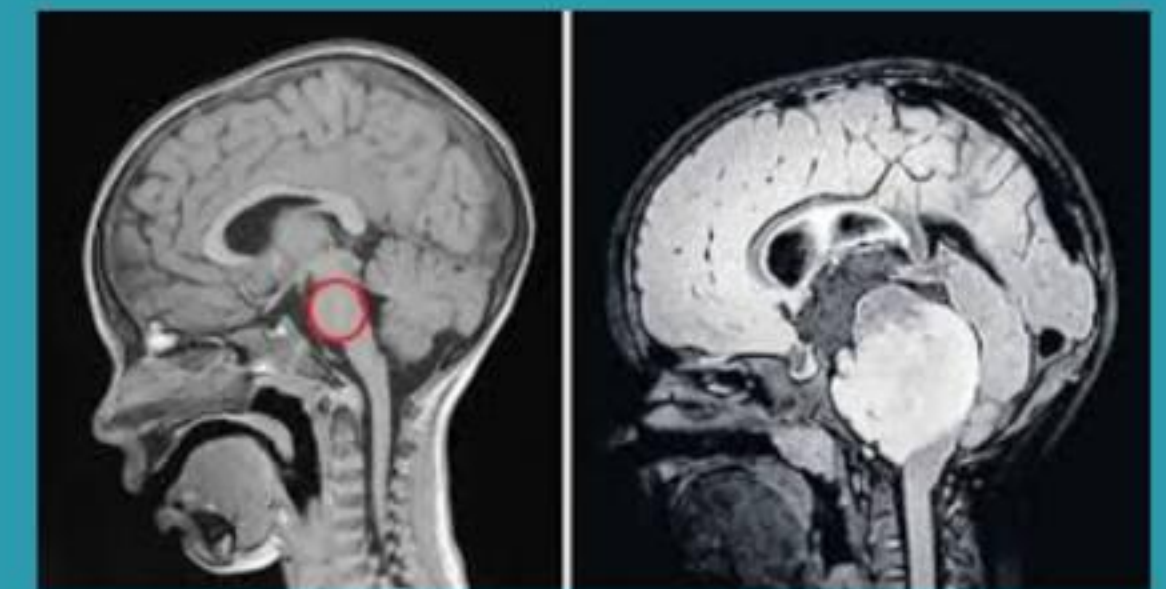
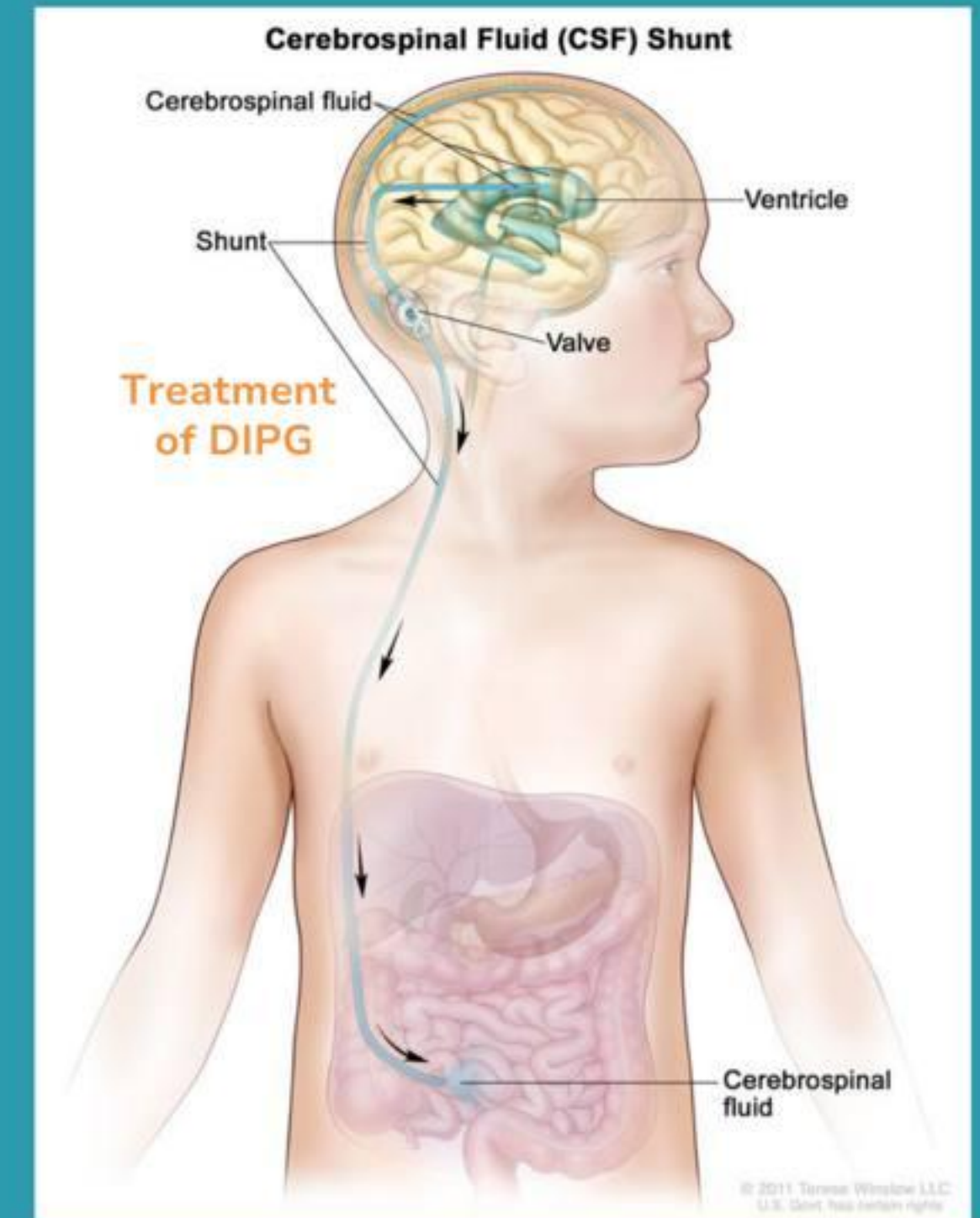
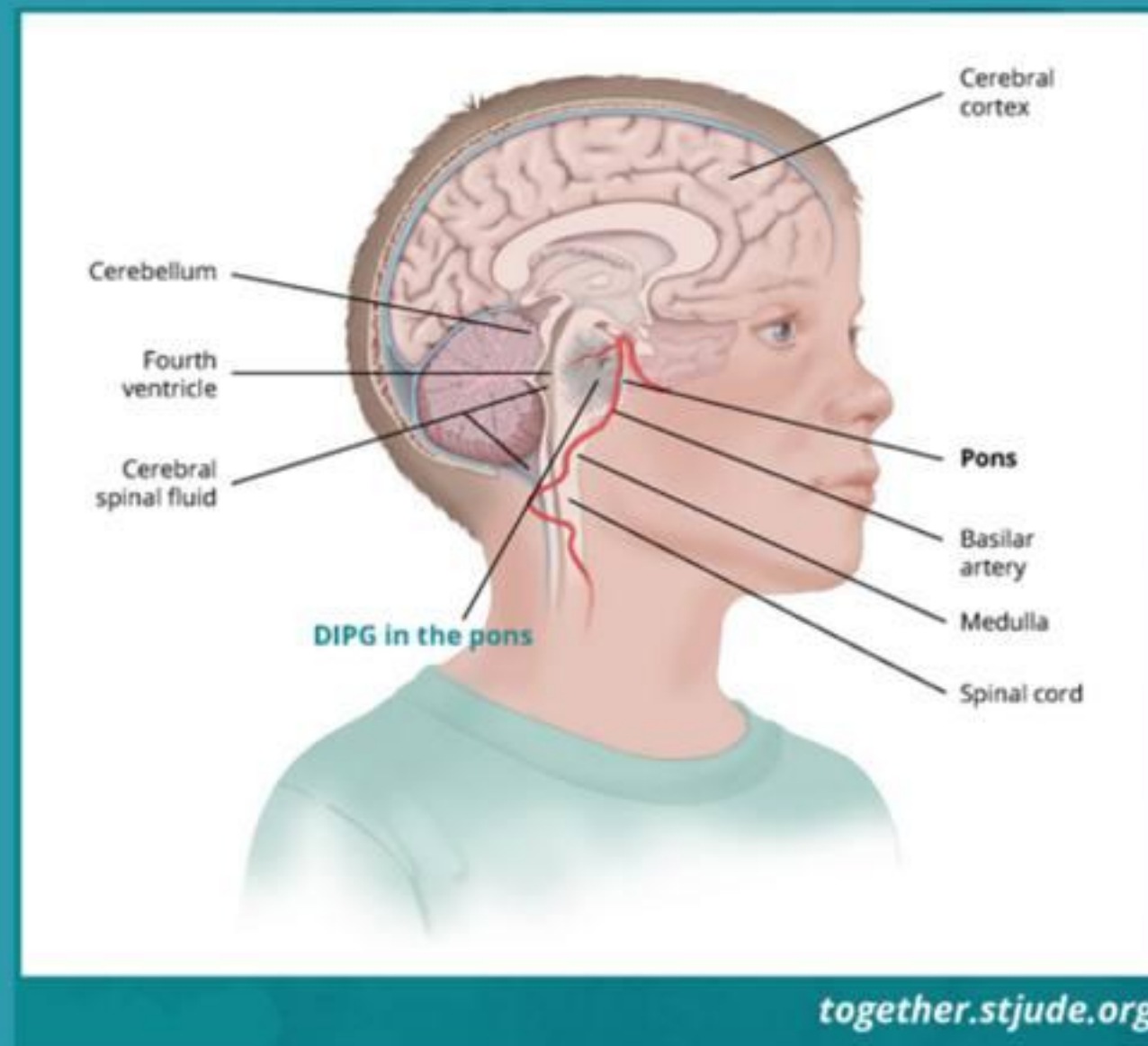
AT/RT: Atypical Teratoid Rhabdoid Tumor

Most AT/RTs are caused by changes in a gene known as **SMARCB1** (also called **INI1**), and less frequently by mutations in a gene called **SMARCA4**. SMARCB1 normally signals proteins to stop tumor growth. But, in AT/RTs, SMARCB1 doesn't function properly and tumor growth is uncontrolled. In addition to occurring in the tumor's DNA, SMARCB1 and SMARCA4 can also be found in a person's own DNA.

AT/RT: Atypical Teratoid Rhabdoid Tumor

- Cell of Origin for All Groups:
 - Not definitively known
 - Believed to arise from embryonal cells, which are capable of differentiating into various cell types
- Anatomical Regions of Origin:
 - Can develop anywhere in the CNS
 - **Brain:** Cerebral hemispheres, ventricles, suprasellar region, pineal gland, cerebellum, and brainstem.
 - **Spinal Cord:** Less common, could originate here

DIPG: Diffuse Intrinsic Pontine Glioma



DIPG: Diffuse Intrinsic Pontine Glioma

- Cell of Origin: glial cells in the brainstem, specifically the pons
- Incidence: US - ~300 children diagnosed per year.
 - DIPG primarily affects children of 5 and 10 years
 - Can occur in younger kids and teens
 - DIPG is rare in adults
- Survival Rates:
 - The 2-year survival rate is very low, and long-term survival is rare

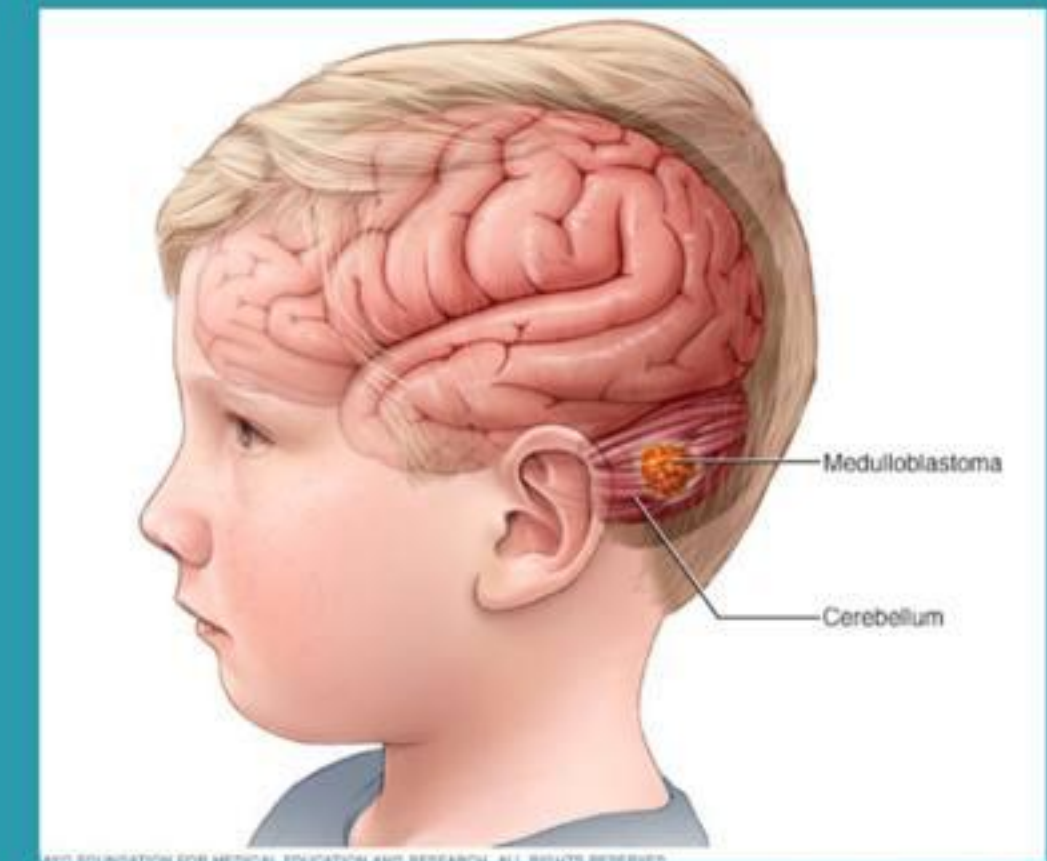
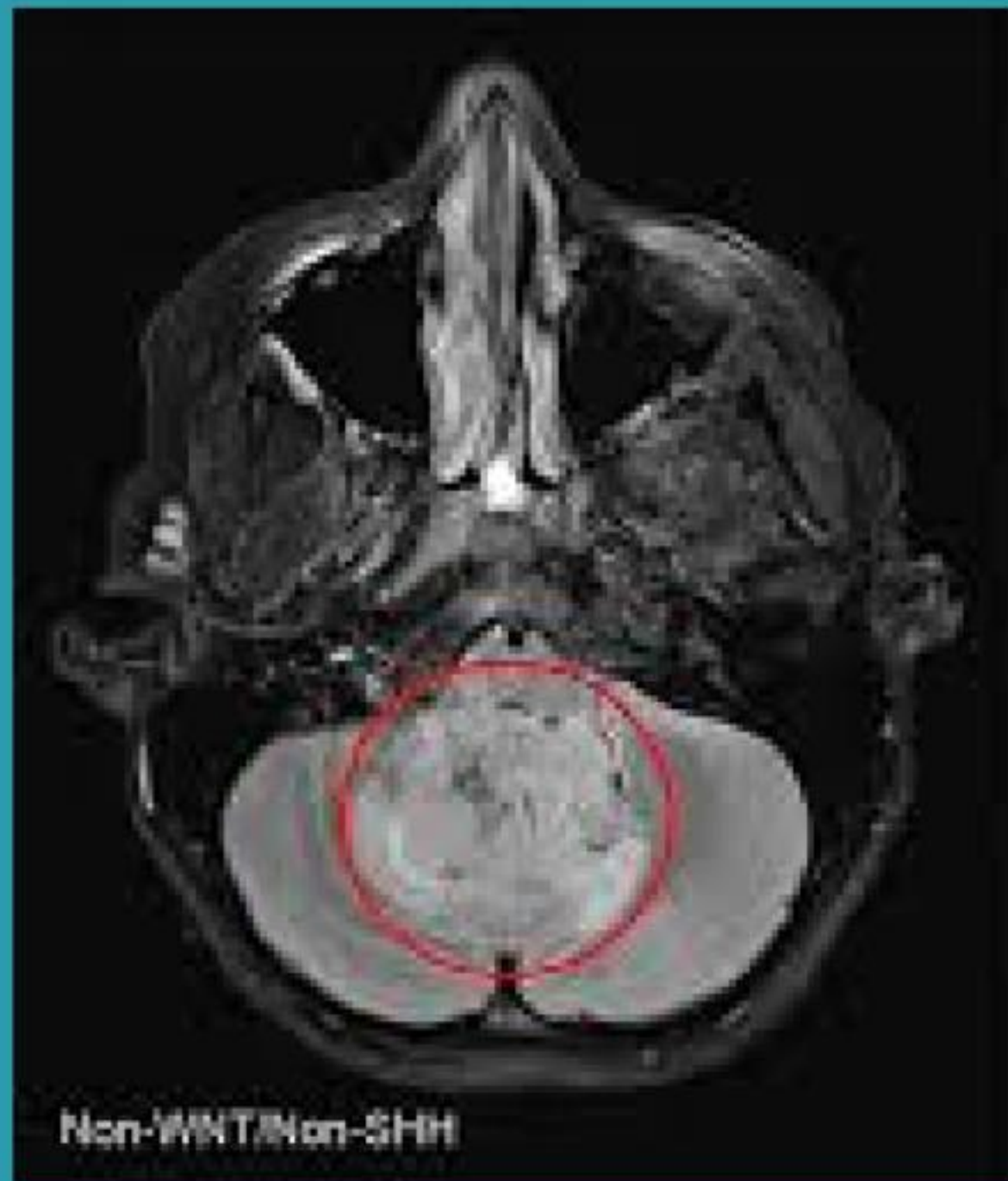
DIPG: Diffuse Intrinsic Pontine Glioma

- Identified Subgroups:
 - H3 K27M-mutant DIPG
 - Characterized by a specific mutation in the histone H3 protein
 - ACVR1-mutant DIPG
 - Mutations in the ACVR1 gene, which is involved in bone morphogenetic protein signaling
 - Other genetic alterations include changes in the TP53, PPM1D, and PDGFRA genes

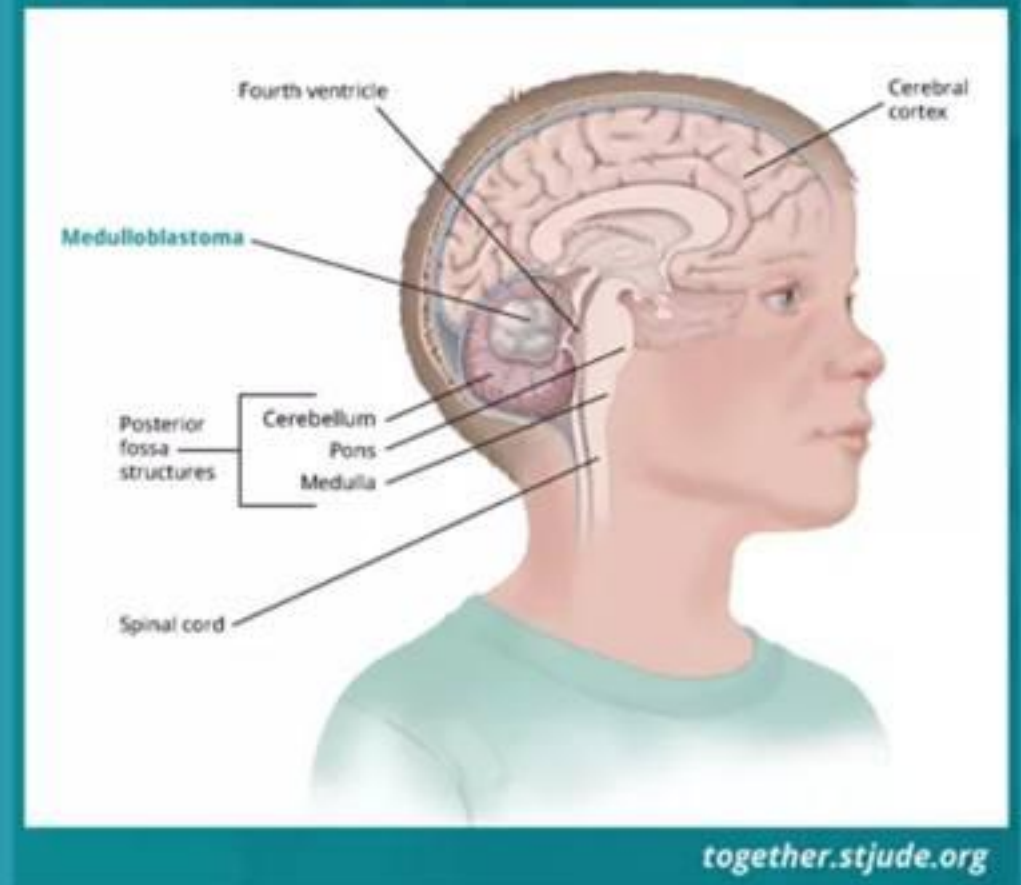
DIPG: Diffuse Intrinsic Pontine Glioma

- Identification of Subgroups:
 - Subgroups identified through whole-genome sequencing, RNA sequencing, and methylation profiling
 - Analyze the genetic and epigenetic aspects of DIPG tumors
- Cell of Origin for All Groups:
 - Generally believed that all DIPG subgroups originate from glial cells in the brainstem

Medulloblastoma



Medulloblastoma



MB: Medulloblastoma

- Medulloblastoma is a type of brain tumor that originates in the cerebellum
- The most common malignant brain tumor in children
- Incidence:
 - 15% - 20% of all pediatric brain tumors
 - Primarily affects children ages 3 to 8
 - Male to female incidence ratio 1.5:1
- Survival:
 - Standard-risk (non-metastatic): 80%-90%
 - High-risk (metastatic/residual tumor): 50%-70%

MB: Medulloblastoma

- Cells of Origin:
 - Granule Neuron Progenitors: Disruption in the signaling pathways that regulate GNP proliferation, such as Sonic Hedgehog pathway, can lead to medulloblastoma
 - Neural Stem Cells: Give rise to neurons and glia. Mutations in proliferation and differentiation can lead to medulloblastoma

MB: Medulloblastoma

- Identified of Subgroups:
 - WNT (Wingless) subgroup: associated with mutations in CTNNB1 gene
 - SHH (Sonic Hedgehog) subgroup: associated with mutations in ELP1 gene
 - Group 3: characterized by mutation in KBTBD4
 - Group 4: most common subgroup, characterized by mutation in KBTBD4

MB: Medulloblastoma

- Identification of Subgroups:
 - Gene Expression Profiling: comparing gene expression profiles from different medulloblastoma samples to identify distinct patterns that correspond to different subgroups
 - DNA Methylation Profiling: analyzing differences in methylation patterns of DNA that can regulate gene expression.

MB: Medulloblastoma

- Cells of origin for each subgroup:
 - WNT: progenitor cells in lower rhombic lip or dorsal brainstem
 - SHH: granule neuron precursors in the external granule layer of cerebellum
 - Group 3: less well-defined, may arise from neural stem cells in brainstem
 - Group 4: less well-defined, may arise from unipolar brush cells or other progenitors in brainstem

Machine Learning Evaluation

- F1 Score:
 - Used to evaluate performance of Machine Learning models
 - Macro F1 score: average f1 score from each cell type
 - Micro F1 score: calculates using total true positives, false positives, and false negatives
- Matplotlib:
 - Visualize performance of different models side by side